OPPORTUNITY BRIEF · SALES AND PRESALES

LPWAN-based condition monitoring for bioenergy plants to track fuel quality and conveyor belt health, reducing unplanned downtime and optimizing fuel handling.

Energy × Energy optimisation × LPWAN (LoRaWAN, NB-IoT, LTE-M) · OS157 v1 · refreshed 2026-08-18

ATTRACTIVENESS	RIGHT TO WIN	COMPETITION	SERVICEABLE MARKET	HORIZON	PORTFOLIO
52 is the world moving	**54** can we play, can we win	**MEDIUM** 0 named in the evidence	**€171m** partial evidence · per year	**NOW** budgeted procurement within 1...	**L0** 3 signals

Attractiveness and right to win are scored and displayed separately and are never combined (SC-12). Competition is a fourth quantity, not a discount on either.

Evidence gap (SC-13). Orange has 3.0 published reference(s) in Energy, below the threshold of 5. A customer conversation here has no proof point behind it yet.

The opportunity

This opportunity is about deploying LPWAN-based condition monitoring for bioenergy plants to track fuel quality and conveyor belt health, reducing unplanned downtime and optimizing fuel handling. It targets European bioenergy operators who are already investing in equipment upgrades and market consultations. The timing is right because these plants are actively seeking solutions to improve operational efficiency and reliability.

Market opportunity

Method	Total (TAM)	Serviceable (SAM)	Obtainable (SOM)	Confidence
Bottom-up: enterprises x adoption x contract value	**€207m** €32.7m – €3.2bn	**€171m** €27.0m – €2.6bn	**€2.6m** €405k – €39.6m	partial
Observed: tendered contract value, annualised	**€172m** €172m – €460m	**€172m** €172m – €460m	**€2.6m** €2.6m – €6.9m	observed

All figures are annual, in EUR. Ranges compound the uncertainty of each factor below; the base case is the middle column of each.

Bottom-up: enterprises x adoption x contract value

Factor	Value	Basis	Source	As of
Enterprises in Energy (EU27_2020)	5,796 enterprises	observed	Eurostat · sbs_sc_ovw	2024
Enterprises adopting LPWAN (LoRaWAN, NB-IoT, LTE-M)	51.7 % of enterprises (10+ employees)	proxy	Eurostat · isoc_eb_iotn2 · E_IOT1	2021
Annual value of one engagement	€299k per enterprise per year	observed	TED (Tenders Electronic Daily) · ted	2026-05-19..2026-08-07
Size-weighted buyer base	1,341 enterprise-equivalents at large-enterprise engagement value	assumption	config/sizing.yaml · size_class_value_weights	size-2026-08-a
Assumed obtainable share of the serviceable market	1.5 % of SAM	assumption	config/sizing.yaml · obtainable_share	size-2026-08-a

Read this before quoting the number. Adoption rate is a declared proxy. Internet of Things by NACE Rev. 2 activity series E_IOT1 is used as a proxy for LPWAN (LoRaWAN, NB-IoT, LTE-M): IoT adoption as the connectivity demand proxy. Contract values are observed from EU public procurement (TED). For private-sector verticals they are used as a proxy for comparable enterprise deal size: the buying entity differs, the scope of work is comparable. Engagement value is scaled by enterprise size class (10-19: x0.04, 20-49: x0.09, 50-249: x0.35, GE250: x1.0), because the observed contract value comes from large-organisation tenders.

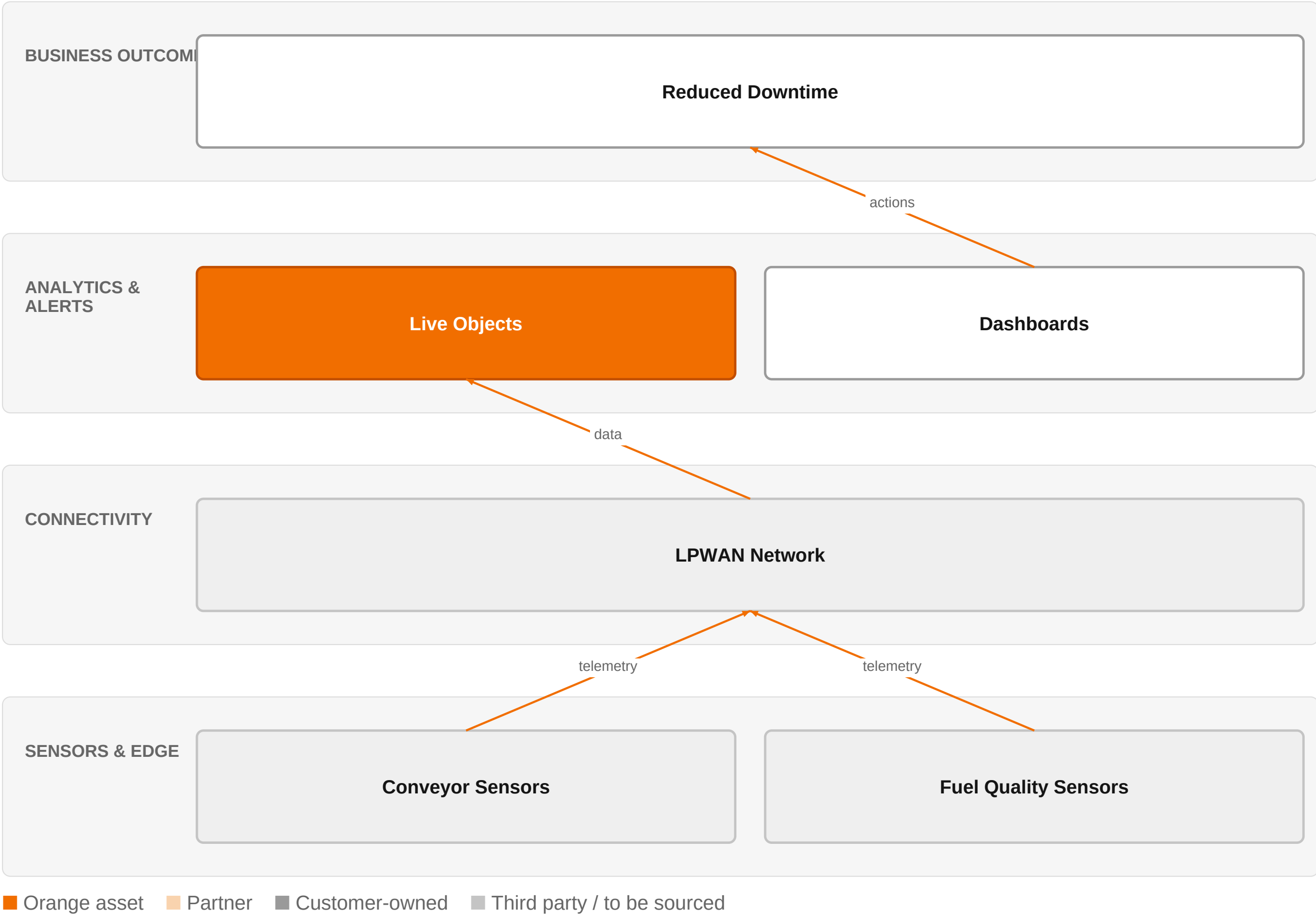
Observed: tendered contract value, annualised

Factor	Value	Basis	Source	As of
Tendered contract value in matching CPV categories	€172m per year	observed	TED (Tenders Electronic Daily) · ted	2026-05-19..2026-08-07
Assumed obtainable share of the serviceable market	1.5 % of SAM	assumption	config/sizing.yaml · obtainable_share	size-2026-08-a

Read this before quoting the number. Observed over 85 days of TED coverage and scaled to a year (x4.3); a short collection window makes that scaling rough. 21 matching notices, matched on vertical via the CPV crosswalk — §4.5.2's warning applies: a crosswalk error lands directly in this number. Public procurement only. Private-sector demand for the same capability is not visible here at all, so this is a floor rather than a market size.

The solution, in one picture

LPWAN Condition Monitoring for Bioenergy



This diagram shows how sensor data flows through LPWAN to Live Objects, enabling analytics that drive operational decisions to reduce downtime.

Boxes marked as Orange assets are validated against the linked business graph; a box the model claimed for Orange without a matching asset is shown as third party.

What has changed

Bioenergy plants are actively investing in equipment upgrades and conducting market consultations for pre-separators and conveyor belts, indicating a focus on improving fuel handling and reducing downtime. The market is seeing increased procurement activity in this area, as evidenced by recent tenders. This creates an opening for condition monitoring solutions that can help optimize these investments.

`Sources: SIG-ED6B4C16F384 ted.europa.eu 2026-07-08; SIG-7A360EFB1A83 ted.europa.eu 2026-05-20`

Who buys, and why now

The buyer is typically the plant operations manager or maintenance director at a bioenergy facility. The pain is unplanned downtime caused by conveyor belt failures or fuel quality issues, which disrupts fuel supply and increases operational costs. The trigger is often a recent breakdown or a planned upgrade cycle where they are already evaluating new equipment and are open to adding monitoring capabilities.

Why it is hot now — every claim bound to a dated source

- Bioenergy plants are investing in equipment upgrades and market consultations for pre-separators and conveyor belts. [SIG-ED6B4C16F384, SIG-7A360EFB1A83]

What Orange would deliver, and why it can win

What Orange would deliver

Orange would deliver a condition monitoring solution using Live Objects as the IoT platform to collect and analyze data from sensors on conveyor belts and fuel quality. The engagement would start with a discovery workshop to understand the plant's specific pain points, then deploy LPWAN sensors (via partners) and integrate them with Live Objects. Orange's IoT experts would configure dashboards and alerts, and Orange Group researchers could help develop predictive models. The solution would be delivered on ISO 27001 certified infrastructure, with SecNumCloud available for sovereign requirements.

Why Orange can win

Orange can win because it combines a robust IoT platform (Live Objects) with deep expertise from its IoT experts and researchers, and offers certified security (ISO 27001, SecNumCloud) that is critical for critical infrastructure. The ability to provide a sovereign, secure solution differentiates Orange from pure-play LPWAN providers. However, the assets are somewhat thin—there is no specific bioenergy domain expertise listed, so Orange would need to partner for domain knowledge.

Named assets behind this — each individually inspectable (LK-08)

Asset	Type	Link	Why it is linked	Curator
Live Objects	offer	L0	offer addresses the use case AND provides the technology	unconfirmed
ISO 27001	certification	SUP	certification Orange holds is required by this vertical	unconfirmed
SecNumCloud	certification	SUP	certification Orange holds is required by this vertical	unconfirmed
IoT experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Orange Group researchers	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed

Competitive landscape

MEDIUM competitive intensity (score 4.8; 0 of 8 listed players appear in this topic's own evidence)

A real field, including at least one player the customer will already know. Expect to be compared.

Competitors include telecom operators like Telia, Vodafone, and Telefonica Tech, who offer LPWAN connectivity and compete in Smart Industries. Industrial specialists like Actility, Honeywell, Schneider Electric, and Siemens have strong OT expertise and are active in Energy. Global systems integrator Capgemini also plays in this space. The customer will compare Orange's offering against these players, particularly on domain expertise and end-to-end solution capability.

`Sources: SIG-ED6B4C16F384 ted.europa.eu 2026-07-08; SIG-7A360EFB1A83 ted.europa.eu 2026-05-20`

Competitor	Category	Position here	Basis	Evidence
Telia Company	Telecom operator peer	sells LPWAN (LoRaWAN, NB-IoT, LTE-M); competes in OX: Smart Industries	structural	—
Vodafone Business	Telecom operator peer	sells LPWAN (LoRaWAN, NB-IoT, LTE-M); competes in OX: Smart Industries	structural	—
Actility	Industrial / OT specialist	sells LPWAN (LoRaWAN, NB-IoT, LTE-M); competes in OX: Smart Industries	structural	—
Capgemini	Global systems integrator	active in Energy; competes in OX: Smart Industries	structural	—
Telefonica Tech	Telecom operator peer	active in Energy	structural	—
Honeywell	Industrial / OT specialist	active in Energy; competes in OX: Smart Industries	structural	—
Schneider Electric	Industrial / OT specialist	active in Energy; competes in OX: Smart Industries	structural	—
Siemens	Industrial / OT specialist	active in Energy; competes in OX: Smart Industries	structural	—

evidenced means this topic's own sources name the competitor — the signal ids are in the last column and are dated and clickable in the radar. **structural** means the curated register says they sell this technology into this vertical, which is true but is not proof they are in this deal. Register version competitors-2026-08-a.

Qualifying questions for the first meeting

1. What is your current unplanned downtime due to conveyor belt failures or fuel quality issues?
2. Are you planning any upgrades to your fuel handling equipment in the next 12 months?
3. Do you have existing IoT or condition monitoring systems in place? If so, what platform do you use?
4. What are your requirements for data sovereignty and security for operational data?
5. Who is responsible for maintenance and operations, and how are decisions made about new technology adoption?

Objections you will meet

They will say	You can answer
We already have a maintenance program and don't see the need for additional monitoring.	That's fair—many plants have preventive maintenance. However, condition monitoring can shift you to predictive maintenance, reducing costs and downtime further. Our solution can complement your existing program by providing real-time data on equipment health.
We are concerned about the reliability of LPWAN in our industrial environment.	LPWAN technologies like LoRaWAN and NB-IoT are designed for industrial use and have proven reliability. We can conduct a site survey to ensure coverage and reliability, and we can also offer alternatives like LTE-M if needed.
We already work with a large industrial automation vendor for our control systems.	That's understandable. Our solution is complementary and can integrate with existing systems via APIs. We focus on the IoT layer and can work alongside your current vendors to provide a complete picture.

Risks and unknowns

The main risk is that bioenergy plants may not see condition monitoring as a priority compared to other investments, or may prefer to work with established industrial automation vendors. There is also uncertainty about the technical integration with existing plant systems and the reliability of LPWAN in industrial environments. Additionally, the business case for monitoring conveyor belts may be less compelling than for other assets, and the customer may need to be convinced of the ROI.

Next action, by role (FR-17)

Role	Do this next
Presales / Proposal	Prepare a bid brief for a bioenergy plant's equipment upgrade project, differentiating our approach by highlighting Live Objects' capabilities for LPWAN device management and data integration, and supporting claims with ISO 27001 and SecNumCloud certifications.
Sales	In a conversation with a bioenergy plant operator's operations manager, say: 'We can help you reduce unplanned downtime and optimize fuel handling with a secure, IoT-enabled condition monitoring solution built on our Live Objects platform, which is ISO 27001 and SecNumCloud certified.'
Strategist / Innovator	Commission a deep-dive brief on LPWAN-based condition monitoring for bioenergy plants, focusing on fuel quality tracking and conveyor belt health, leveraging IoT experts and Orange Group researchers, and mapping how Live Objects can underpin the solution.

Evidence

Every claim in this brief resolves to one of these. Tier 1 is an authoritative primary source; tier 4 is vendor-published material, discounted in scoring (§4.3.7).

Signal	Date	Publisher	T	Title and link
SIG-ED6B4C16F384	2026-07-08	ted.europa.eu	1	Netherlands – Heat-exchange units, air-conditioning and refrigerating equipment, and filtering ...
SIG-D85076BA625E	2026-05-28	ted.europa.eu	1	Lithuania – Methane monitoring – (VPP-680) Gamtiniu duju (metano) nuotekiu matavimo, duomenu an...
SIG-7A360EFB1A83	2026-05-20	ted.europa.eu	1	Denmark – Rubber conveyor belts – BIO_Transportband & ruller_26-31_FA

How this brief was made

Computed, not written. Attractiveness, right to win, competitive intensity and every figure in the market-opportunity section are computed from stored inputs and can be re-derived (DR-05, NFR-01). No language model produced a number anywhere in this document (§4.4.4).

Written, then checked. The narrative sections and the solution diagram were generated from this topic's evidence and from the named asset and competitor lists. Factual sections must cite signal ids attached to this topic; sections that failed that check, or that contained a generated quantity or an unsupplied organisation, were removed rather than rewritten.

What	Version	When
Scores — weight set	w-2026-08-a	2026-08-18T19:08:26+00:00
Market sizing — assumptions	size-2026-08-a	2026-08-18T21:06:16+00:00
Competitor register	competitors-2026-08-a	2026-08-18T21:06:19+00:00
Narrative — prompt / model	describe-v1 / deepseek-chat	2026-08-19T04:57:28+00:00
Pipeline	0.1.0	2026-08-18

Generated 2026-08-19 04:58 UTC. Scores computed under a different weight set are not comparable with these (SC-10), and neither are market sizes computed under a different sizing version.